

Limits (Part IV)

Find the values of the limits algebraically.

$$\textcircled{1} \lim_{b \rightarrow \frac{1}{3}} \frac{27b^3 - 1}{3b - 1}$$

$$\textcircled{2} \lim_{t \rightarrow -7} \frac{t + 7}{t^3 + 343}$$

$$\textcircled{3} \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\tan \theta}$$

$$\textcircled{4} \lim_{A \rightarrow \frac{\pi}{2}} \frac{\cot A}{\cos A}$$

$$\textcircled{5} \lim_{B \rightarrow 0} \frac{1 - \cos B}{\sin^2 B}$$

Use one of the Pythagorean identities.

$$\textcircled{6} \lim_{\beta \rightarrow \pi} \frac{\tan^2 \beta}{\sec \beta + 1}$$

$$\textcircled{7} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin(2\theta)}{\cos \theta}$$

$$\textcircled{8} \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\sin^2 x + \cos(2x)}$$

$$\textcircled{9} \lim_{t \rightarrow 0} \frac{\sin t \tan t}{\tan(-t)}$$

$$\textcircled{10} \lim_{w \rightarrow \frac{3\pi}{2}} \frac{\sec(-w) \sin w}{\sec w}$$

Find the value of each limit. Write the number of the limit law used on each step.

$$\textcircled{11} \lim_{x \rightarrow 0} \frac{x \log_2(x+2)}{\sin x}$$

$$\textcircled{12} \lim_{x \rightarrow 0} \frac{\sqrt{x} \cos^2 x - \sqrt{x}}{x}$$

$$\textcircled{13} \lim_{x \rightarrow 0} \frac{\sin^4 x}{x^4}$$

$$\textcircled{14} \lim_{\theta \rightarrow 0} \frac{\sin \theta + \theta}{\theta}$$

Start by dividing both sides of the fraction by "x".

$$\textcircled{15} \lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(7x)}$$

$$\textcircled{16} \lim_{y \rightarrow 0} \frac{\tan y}{y}$$

$$(17) \lim_{\theta \rightarrow 0} [\theta \cot(7\theta)]$$

Homework Answers

$$(1) \boxed{3}$$

$$(2) \boxed{\frac{1}{147}}$$

$$(3) \boxed{1}$$

$$(4) \boxed{1}$$

$$(5) \boxed{\frac{1}{2}}$$

$$(6) \boxed{-2}$$

$$(7) \boxed{2}$$

$$(8) \boxed{\frac{1}{2}}$$

$$(9) \boxed{0}$$

$$(10) \boxed{-1}$$

$$(11) \text{ 3, 4, 6, 8 } \boxed{1}$$

$$(12) \text{ 3, 8 } \boxed{0}$$

$$(13) \text{ 7 } \boxed{1}$$

$$(14) \text{ 1, 6 } \boxed{2}$$

$$(15) \text{ 4 } \boxed{\frac{3}{7}}$$

$$(16) \text{ 3 } \boxed{1}$$

$$(17) \text{ 3, 4, 6, 8 } \boxed{\frac{1}{7}}$$